

**ECON 101—Introduction to Microeconomics
Spring 2023**

Midterm I

February 21, 2023

Multiple choice. Please write your answers here!

1	2	3	4	5	6	7	8	9	10

Do not write below this area!

MCQ	Problem 1	Problem 2	Problem 3	Total

Part I: Multiple Choice

Please record all answers in the front sheet. Each question is worth 2 points for a total of 20 points.

Version	I	II	III	IV
Question				
1	(d)	(d)	(a)	(d)
2	(c)	(d)	(c)	(a)
3	(b)	(c)	(c)	(a)
4	(d)	(c)	(b)	(b)
5	(c)	(d)	(a)	(a)
6	(c)	(a)	(d)	(b)
7	(c)	(d)	(c)	(d)
8	(a)	(b)	(b)	(d)
9	(a)	(b)	(a)	(d)
10	(c)	(b)	(b)	(c)

Part II: Long Problems

Please answer *only* in the space provided. Be neat and give complete arguments. You can use as much scratch paper as you need but your scratch work will not be considered.

Problem 1

A market analysis show that there are two types of consumers for coffee at Nazarbayev University. The first type are mild consumers of coffee and have an inverse demand curve of $P = 900 - 90Q$. The second type have a *caffeine addiction* and have an inverse demand curve of $P = 1200 - 60Q$. Quantities are in cups of coffee and the price is in tenge. In Nazarbayev University there are N_1 individuals who are mild consumers, and N_2 consumers with a *caffeine addiction*, where:

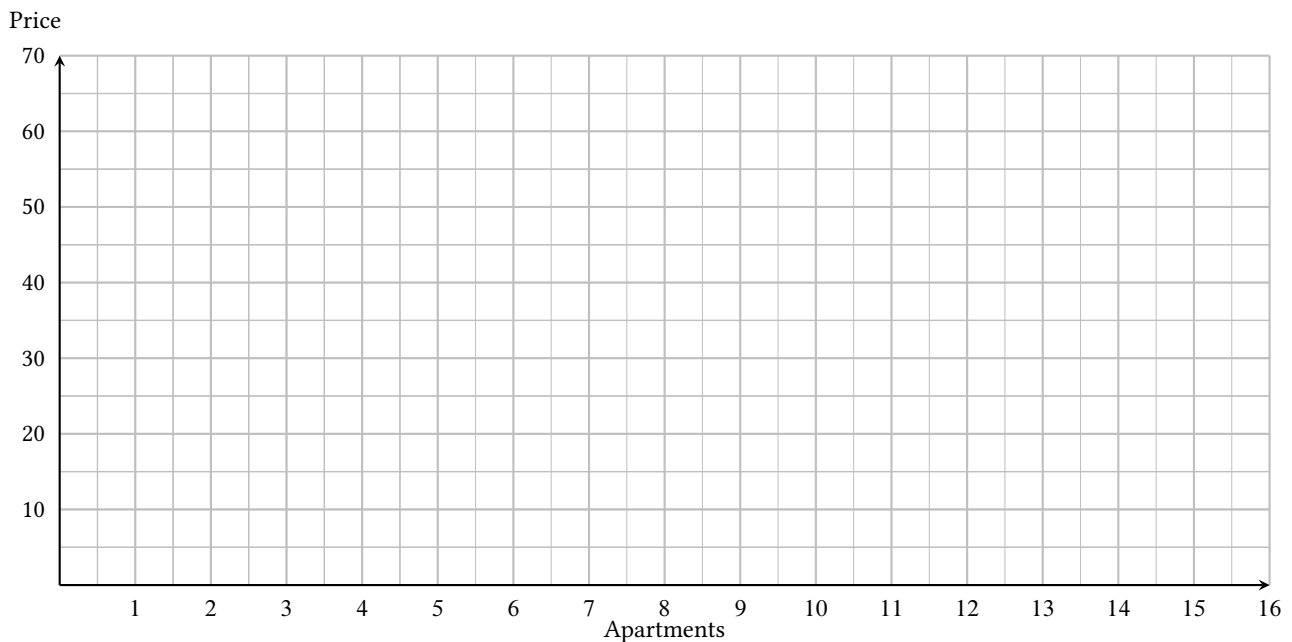
- (a) (5 points) Using the demand curves above, derive the market demand for coffee in Nazarbayev University. Plot the market demand curve.

- (b) (10 points) Suppose that the market supply for coffee in Nazarbayev University is given by $Q^S = 10P$. Find the equilibrium price and quantity. How much does each individual consumer buy at the equilibrium price?
Hint: Check the equilibrium price and quantity with your graph.

Problem 2

The reservation value or reservation price of a consumer is the price at which she is just indifferent between renting or not renting an apartment. At any price below the reservation price the consumer will demand one apartment, at any price above the reservation price the consumer will demand zero apartments, and exactly at the reservation price the consumer will be indifferent between having zero or one apartment.

In this problem you will consider one tiny part of the market for apartments in Astana. Due to the situation with Russia, the Astana housing market recently experienced an influx of Russian immigrants seeking to find housing within the city. Suppose that we have 8 Kazakh nationals and 8 Russian immigrants who want to rent an apartment. Their reservation prices are given below. Kazakh nationals are labeled with uppercase letters and the Russian immigrants with lowercase letters.



- (a) (5 points) Plot the domestic demand and immigrant demand curves in the graph above. Make sure to distinguish them by labeling them properly. *Hint: The demands will have a “staircase” shape. Arrange the numbers and identities first by decreasing reservation price and then alphabetically.*
- (b) (5 points) What is the main difference that you see between the domestic and immigrant demand curves?
- (c) (5 points) Plot the aggregate demand for apartments in Astana in the graph above. Label it properly to distinguish it from the other two. *Hint: This is really no different from what you have done above. Arrange all numbers first by decreasing reservation price, then by nationality (first Kazakh then Russian), and finally alphabetically.*
- (d) (5 points) Consider only the domestic demand. Suppose that if consumers are indifferent between renting or not, i.e., when their reservation value equals the rental price, they choose to rent. Suppose that the supply of apartments is fixed at 5 units. What is the highest price that would make the domestic demand for apartments equal to 5 units?

The highest price would be _____.

What is the lowest price that would make the domestic demand for apartments equal to 5 units?

The lowest price would be _____.

- (e) (5 points) Suppose that apartment owners have all the bargaining power and can charge the highest possible market price. Consider the domestic market only. Which of the A–H people end up getting apartments? What is their consumer surplus?

- (f) (5 points) Now consider the aggregate demand. Continue assuming that if consumers are indifferent between renting or not, they choose to rent. Suppose, as in (c), that if renters of different nationalities have the same reservation price, the Kazakh one comes first. What is the highest price that would make the aggregate demand for apartments equal to 5 units?

The highest price would be _____.

What is the lowest price that would make the aggregate demand for apartments equal to 5 units?

The lowest price would be _____.

- (g) (5 points) Again, suppose that apartment owners can charge the highest possible market price. Which people among both locals and immigrants end up getting apartments?

- (h) (5 points) Consider the highest possible market price you found in (f) and the identifies of the *Astana* buyers you found in (g). What is the domestic consumer surplus?

- (i) (5 points) Consider the aggregate demand and domestic consumer surplus In order to help *Astana* renters, would you recommend putting a rent ceiling equal to the highest price you found in (d) and (e)?

Problem 3

Kazakhstan's market for milk is described by the following domestic demand and supply schedules:

$$Q_D^{\text{domestic}} = 10 - \frac{1}{100}P; \quad Q_S^{\text{domestic}} = \frac{P}{100},$$

where the quantities are in millions of liters and the price is in KZT/liter.

The world price of milk, in KZT/liter is currently $p^{\text{world}} = 177.58$.

- (a) (5 points) Consider first the situation of a *free-trade* market equilibrium. What would be the free-market equilibrium price and quantity? What would be the amount imported and the amount domestically produced? Calculate the items asked and also illustrate your answer with a graph.

- (b) (5 points) In an effort to protect national milk producers, the government of Kazakhstan imposes a tariff of $T = 260.82$ KZT/liter. What would be the new market equilibrium after the tariff is imposed? Calculate the new price and quantities and illustrate your answer with a new graph.

- (c) (10 points) Fill the table (just input the numbers, you can do your calculations below):

	Free Trade	Tariff	Change
Consumer Surplus			
Domestic Producer Surplus			
Government Expenditures			
Social Surplus			